



# Aerodynamic Analysis of Concorde and Su-57 Nose Sections Using Schlieren Imaging

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## Team Members

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# Introduction

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long, slender, pointed droop nose designed to minimize wave drag during cruise.



a 3-D chined, stealth-driven geometry enabling high maneuverability at the cost of stronger shock structures

# Objectives

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This laboratory project has two primary objectives:

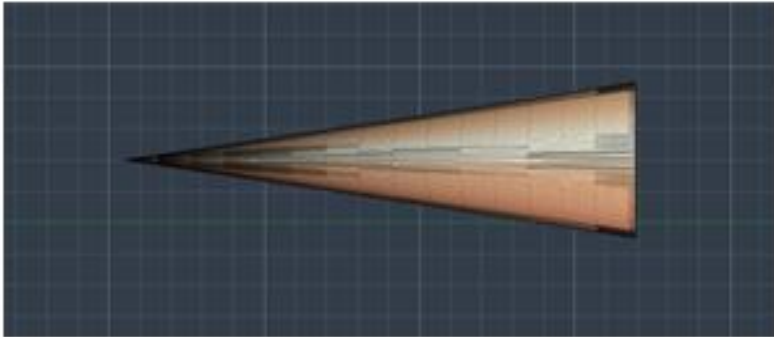
1. **Experiment:** Use Schlieren imaging to capture shock wave geometry over Concorde and Su-57 nose models across a range of angles of attack ( $0^\circ$ – $10^\circ$ ) at Mach 2.2.
2. **Theoretical validation:** Compare experimental shock angles with predictions from:
  - Oblique shock relations ( $\theta - \beta - M$  charts) for 2D wedge baseline
  - Taylor-Maccoll conical flow theory for 3D axisymmetric Concorde nose

# Model fabrication

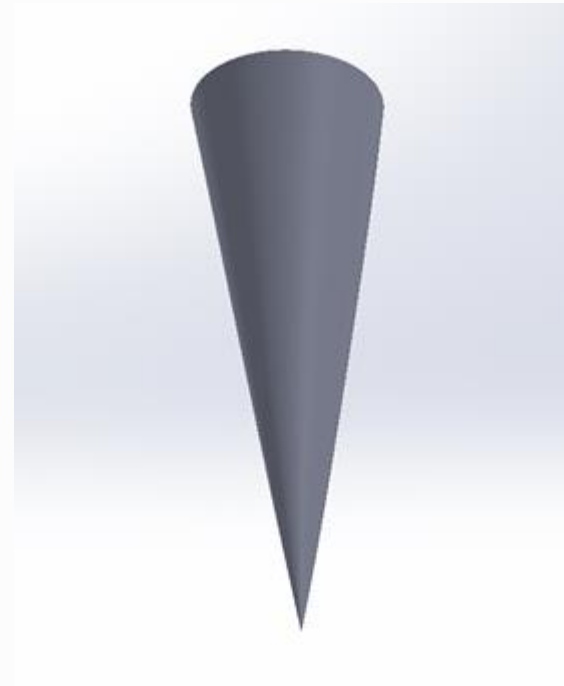
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# CAD Design

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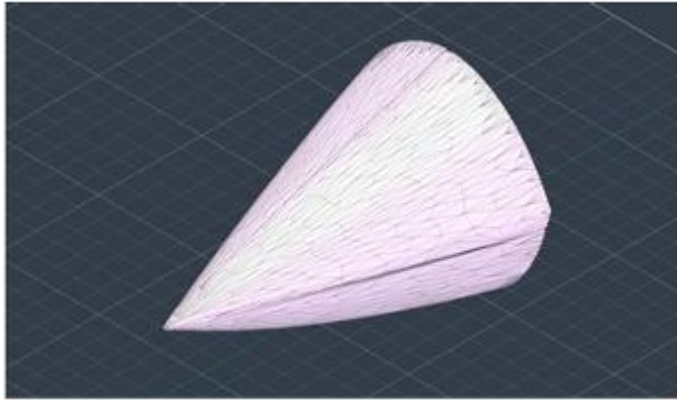


- Height of the model- 85 mm
- Maximum diameter- 25 mm

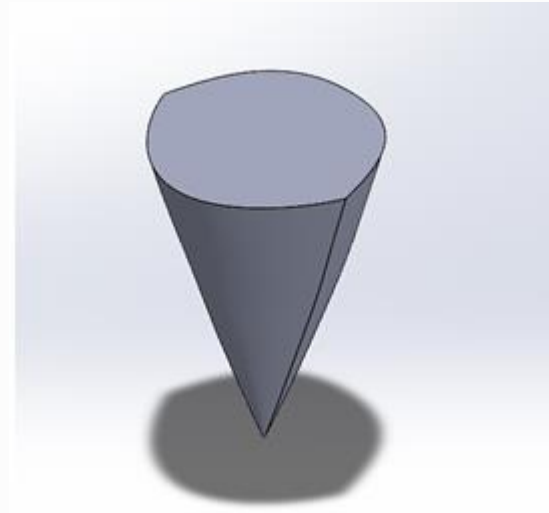


# CAD Design

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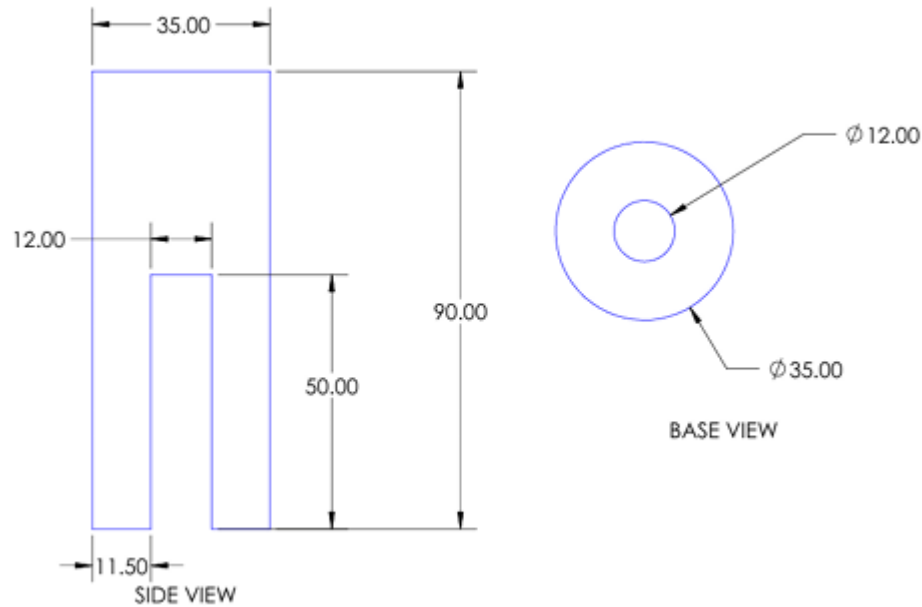


- Height of the model- 42.5 mm
- Maximum diameter- 30 mm



# Material Preparation

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Mild steel was selected as the material for the strut due to its ability to withstand the high pressure loads encountered in the high-speed wind tunnel without undergoing deformation.

A mild steel beam matching the required maximum diameter was chosen and cut to the specified height.

# Material Preparation

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# CNC machining and EDM

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(a) Electrical Discharge Machine



(b) EDM Specimen setup

# Welding

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For the attachment of the supporting rod to the finished specimen, we had two options: brass welding and threading. Since the CNC holding position buffers around 10 mm, the drill was not sufficient to make the threading process we had to do brass welding

# other

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# Final model

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# Experiment

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# Wind tunnel

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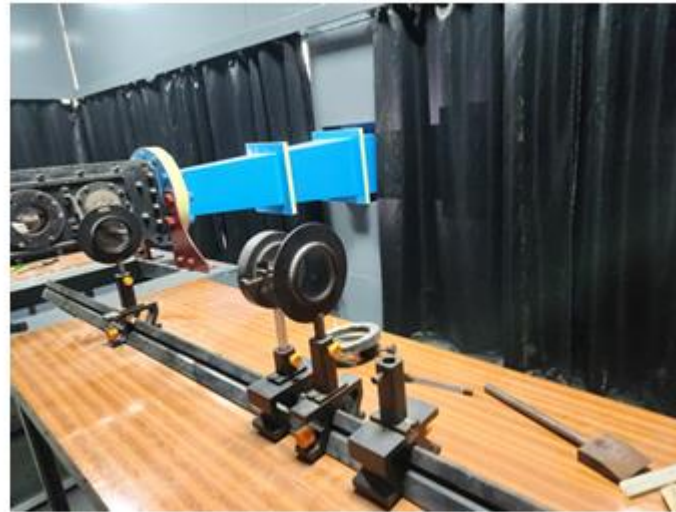


- blowdown-type supersonic wind tunnel
- designed to operate at a Mach number of  $M = 2.2$ .
- The settling chamber supplies air at an external pressure of 6 atm.
- The test section has a cross-sectional area of  $100 \text{ mm} \times 50 \text{ mm}$



# Schlieren Setup

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Schlieren imaging is an optical flow visualization technique developed in 1864 by August Toepler to observe fluid flows by detecting spatial variations in refractive index

# Schlieren Setup

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When light passes through a medium with density gradients (as in a shock wave), the local refractive index changes, causing the light rays to bend (refract). Schlieren imaging converts these deflections into visible brightness differences, which help us visualize the density gradient of the flow. Shock waves produce very sharp density gradients, which deflect light rays.



# Schlieren Setup

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# Schlieren

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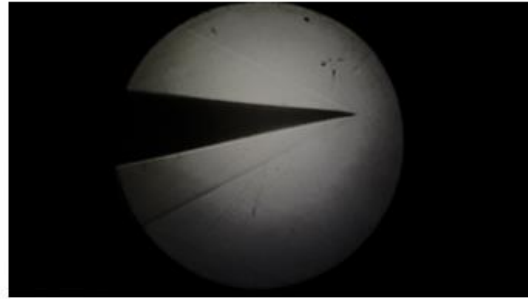


# Results

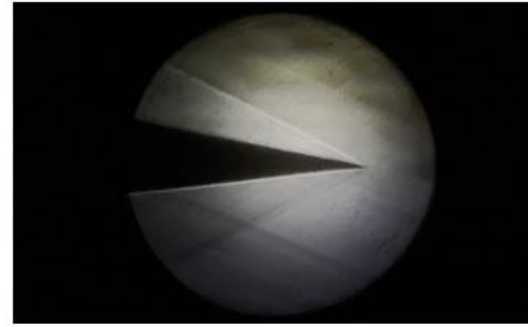
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# Experimental images

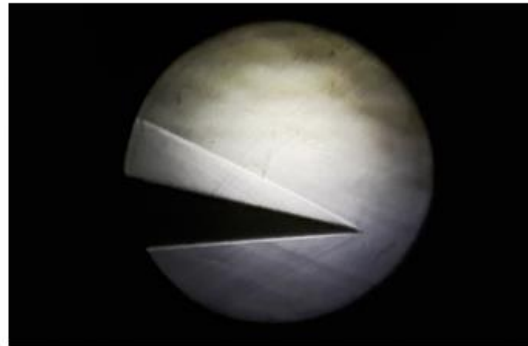
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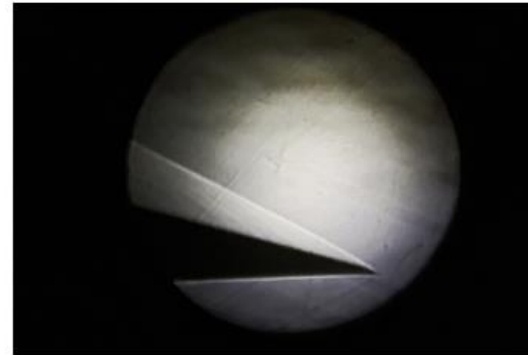
(a)  $0^\circ$  AoA



(b)  $5^\circ$  AoA



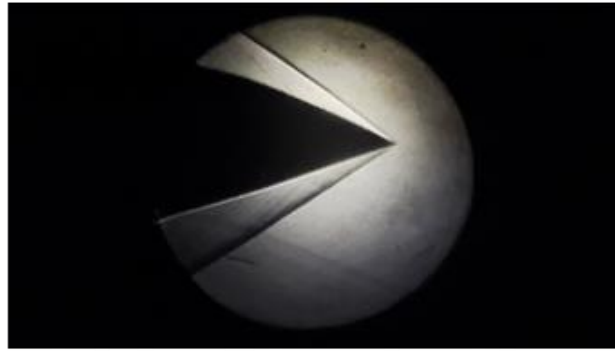
(c)  $8^\circ$  AoA



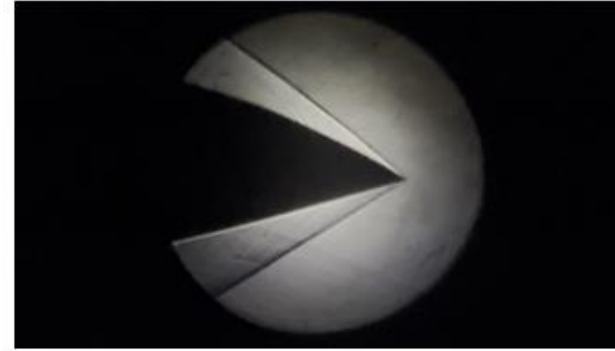
(d)  $10^\circ$  AoA

# experimental images

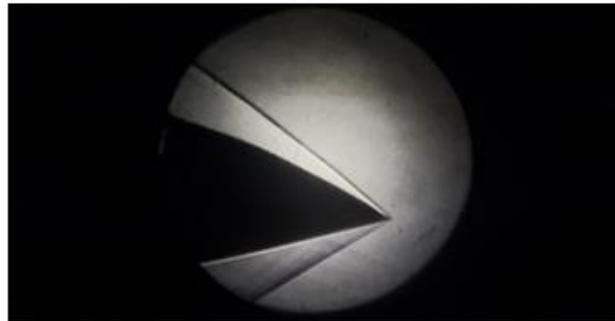
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(a)  $0^\circ$  AoA



(b)  $5^\circ$  AoA



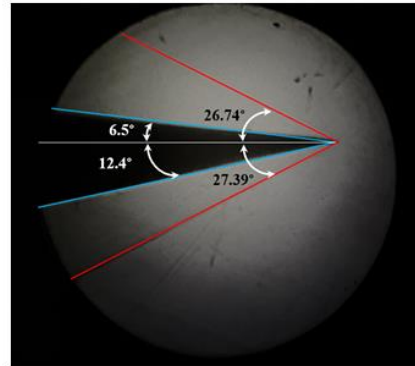
(c)  $8^\circ$  AoA



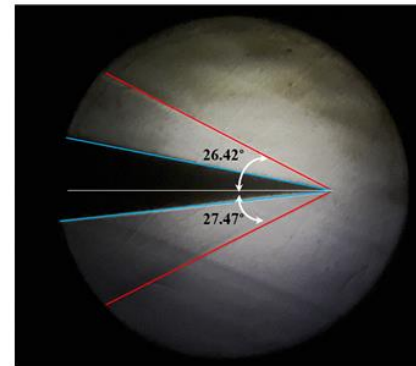
(d)  $10^\circ$  AoA

# Shock Angle measurement

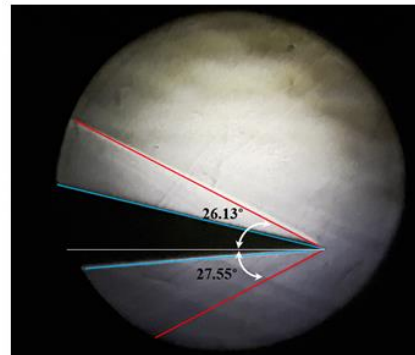
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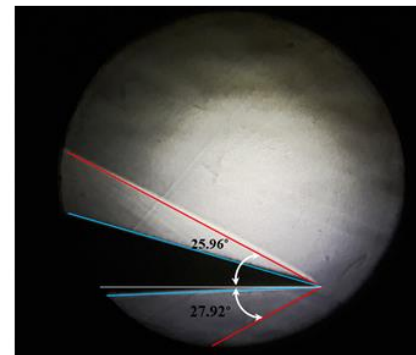
(a) 0° AoA



(b) 5° AoA

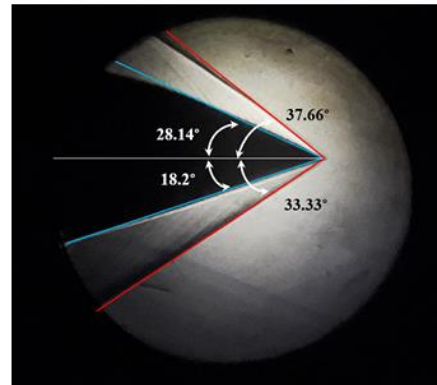


(c) 8° AoA

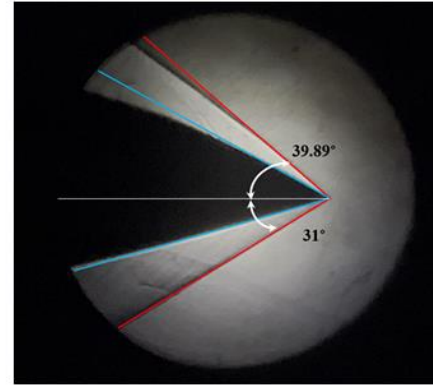


(d) 10° AoA

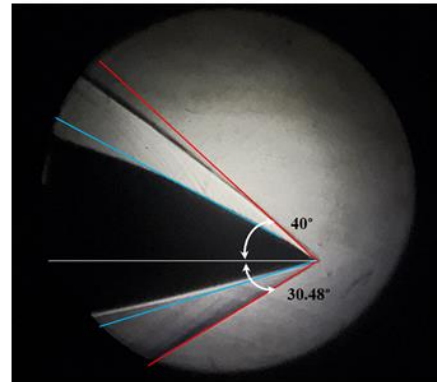
# Shock Angle measurement



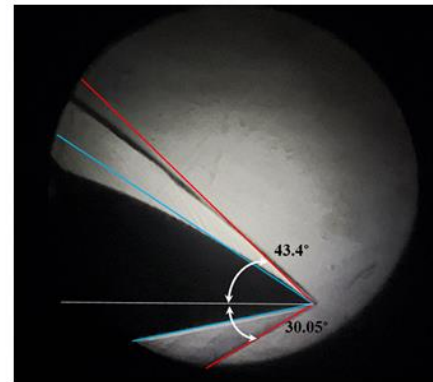
(a) 0° AoA



(b) 5° AoA

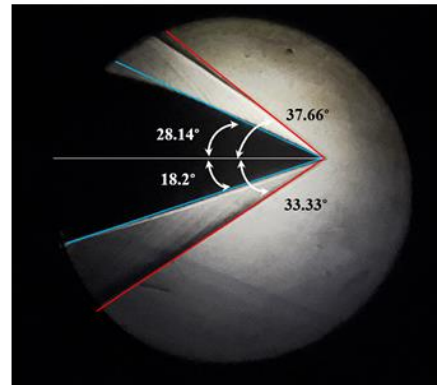


(c) 8° AoA

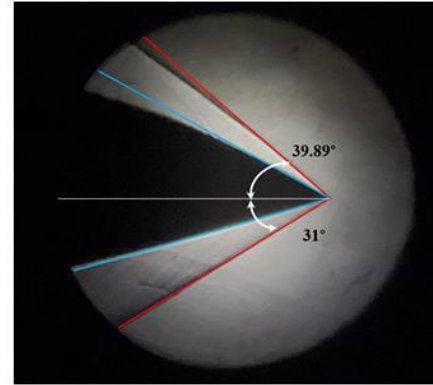


(d) 10° AoA

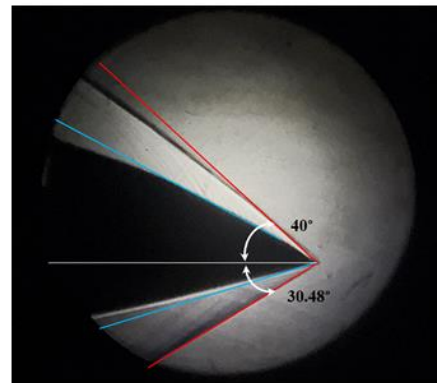
# Shock Angle measurement



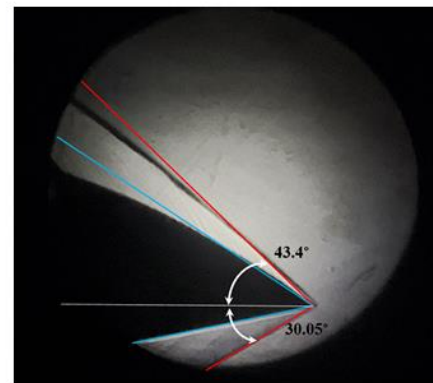
(a) 0° AoA



(b) 5° AoA



(c) 8° AoA



(d) 10° AoA



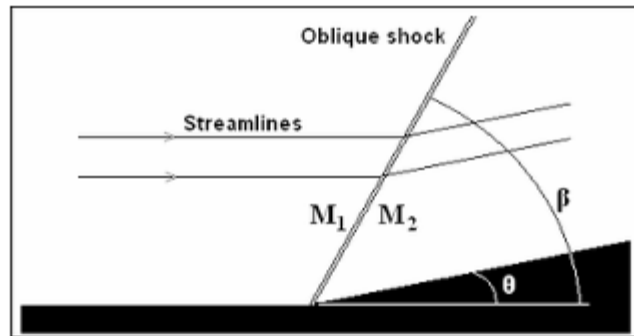
# Theoretical Analysis

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# 2D Wedge Theory

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$$\tan \theta = 2 \cot \beta \left( \frac{M_1^2 \sin^2 \beta - 1}{M_1^2 (\gamma + \cos 2\beta) + 2} \right)$$

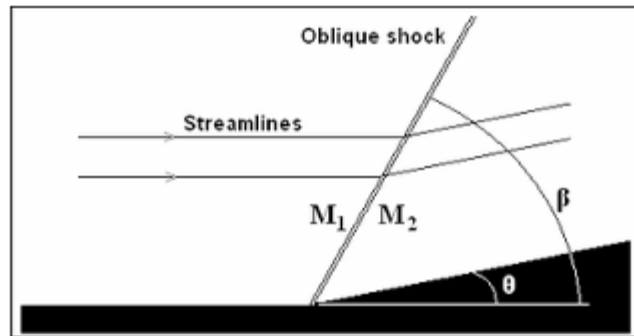


- The Concorde nose can be approximated as a two-stage wedge with flow deflection angles of  $6.5^\circ$  and  $12.4^\circ$ .
- From the  $\theta$ – $\beta$ – $M$  relation for Mach 2.2 we get  $\beta'_1 = 32^\circ$  and  $\beta'_2 = 37.87^\circ$ .
- From the experiment image, the corresponding oblique shock angles were found to be  $\beta_1 = 26.74^\circ$  and  $\beta_2 = 27.39^\circ$ . (where,  $\beta_1$  is for lower surface and  $\beta_2$  for upper surface)

# 2D Wedge Theory

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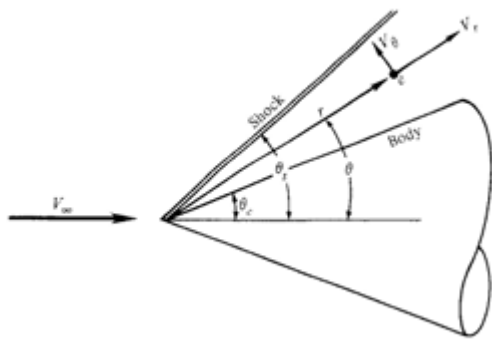
$$\tan \theta = 2 \cot \beta \left( \frac{M_1^2 \sin^2 \beta - 1}{M_1^2 (\gamma + \cos 2\beta) + 2} \right)$$



- We can see that for a 2D wedge, for the same Mach number and same deflection, we have a strong oblique shock, with large  $\beta$ . The variation from the experimental data is  $5^\circ$ -  $10^\circ$ . This is due to the 3D relieving effect, where a 3-D body generates weaker shocks (smaller pressure jumps, smaller drag) than a 2-D body with the same local deflection angle, because the flow can “escape” sideways around the body.

# Taylor Macolle

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The main assumptions are:

- Flow is steady, inviscid, and adiabatic (perfect gas, typically  $\gamma = 1.4$ ).
- The geometry is a right circular cone with semi-vertex (half) angle  $\theta_c$ .
- Axisymmetric, conical flow: all flow properties are constant along rays from the apex (depend only on the polar angle, not on radius or azimuth).
- A single attached conical shock exists whose apex coincides with the cone apex.

# Taylor Macolle

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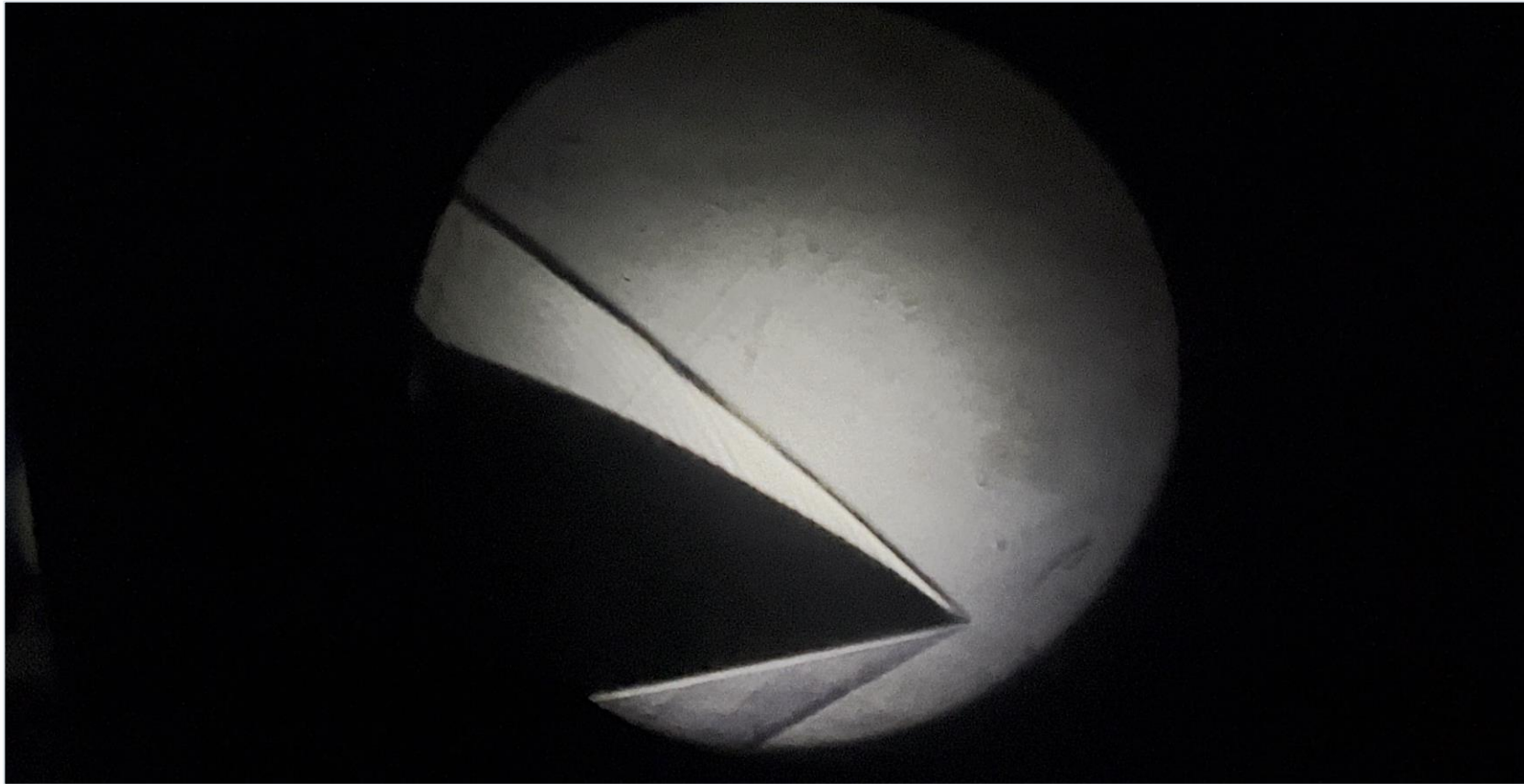
The real Concorde nose is an ogive (non-linear) profile rather than a perfect straight line cone, but it is long, slender, and nearly rotationally symmetric about the aircraft axis.

The Concorde nose can therefore be approximated as a right circular cone of semi-vertex angle  $12^\circ$ , obtained from the CAD nose profile.

Under the assumptions of inviscid, axisymmetric, conical flow at zero incidence, the external flow is governed by the Taylor–Maccoll equation, from which the attached shock angle is determined numerically using a MATLAB solver to be  $\beta_{TM} = 27.235691^\circ$ , which is close to the experimentally observed value.

# Boundary layer

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# Boundary layer

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- leeward (top) side - local shock is weaker, imposed adverse pressure gradient is smaller and the boundary layer remains relatively thin
- Across this thin layer, the density changes from a low value at the wall to a higher value in the freestream over a short distance, producing a steep density gradient at the boundary-layer edge.
- schlieren, which is sensitive to density gradients, this sharp gradient appears as a clear bright band.

# Trend Analysis of shock Angle VS AoA

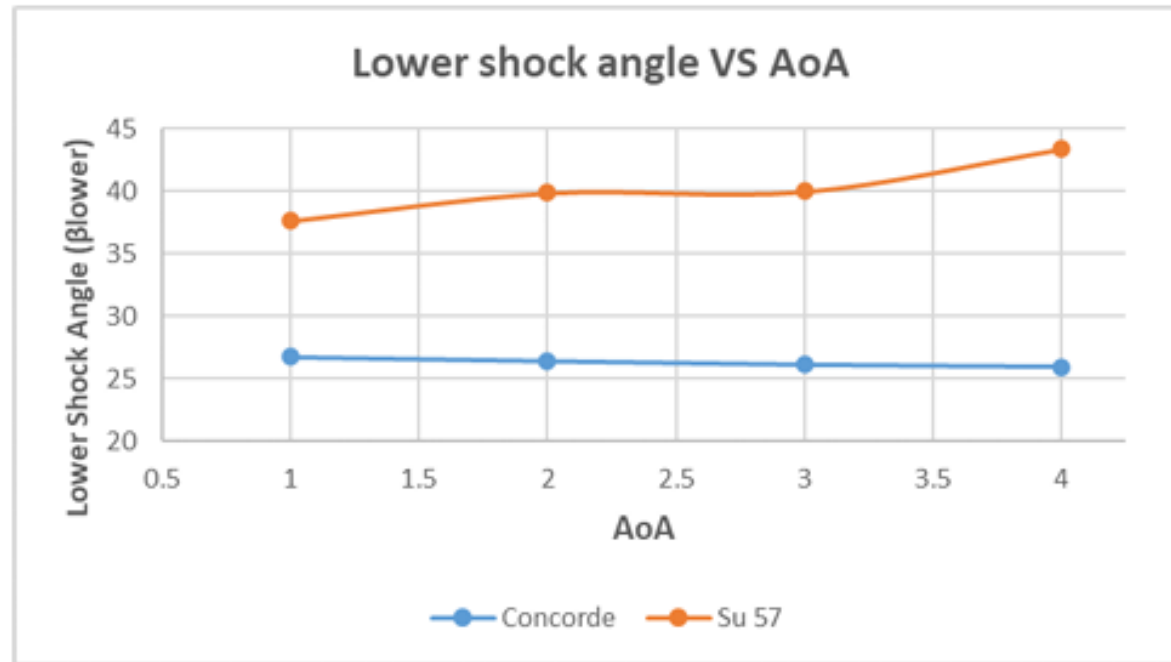
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| Concorde         |                                       |                                       |                       |
|------------------|---------------------------------------|---------------------------------------|-----------------------|
| AoA ( $\alpha$ ) | Lower Shock Angle ( $\beta_{lower}$ ) | Upper Shock Angle ( $\beta_{upper}$ ) | Net Change in $\beta$ |
| 0°               | 26.74                                 | 27.39                                 | N/A                   |
| 5°               | 26.42                                 | 27.47                                 | 0.3                   |
| 8°               | 26.13                                 | 27.55                                 | 0.3                   |
| 10°              | 25.96                                 | 27.92                                 | 0.4                   |
| Su 57            |                                       |                                       |                       |
| AoA ( $\alpha$ ) | Lower Shock Angle ( $\beta_{lower}$ ) | Upper Shock Angle ( $\beta_{upper}$ ) | Net Change in $\beta$ |
| 0°               | 37.66                                 | 33.33                                 | N/A                   |
| 5°               | 39.89                                 | 31                                    | 2.3                   |
| 8°               | 40                                    | 30.48                                 | 0.5                   |
| 10°              | 43.4                                  | 30.05                                 | 3.4                   |



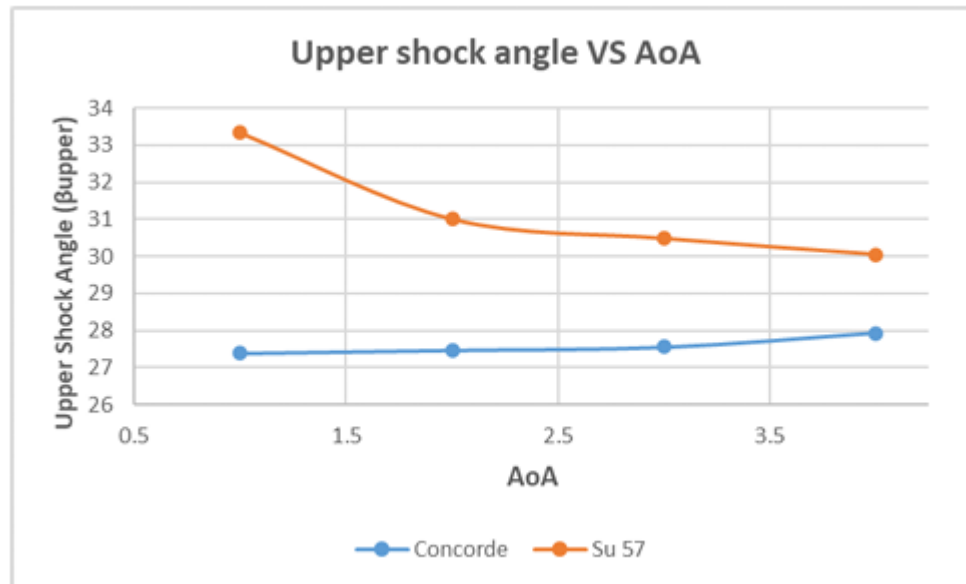
# Trend Analysis of shock Angle VS AoA

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# Trend Analysis of shock Angle VS AoA

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THANK YOU